Task 1.1.3 Big Data: Multi-File Sales Calculation Report

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1. Executive Summary & Key Results

The objective of this task was to process three separate four-month period CSV files for the year 2025 (vendas_2025_cuadrimestre1.csv, vendas_2025_cuadrimestre2.csv, and vendas_2025_cuadrimestre3.csv) to calculate both individual four-month totals and the overall annual total sales volume and revenue.

Results Breakdown

- **Annual Cumulative Totals:** 1131738.68€ and 3450000 total sales.

2. Technical Challenges Encountered

- **Multi-File Dataset Handling:** Unlike previous single-file tasks, the data was split into three separate CSV files representing different periods of the year. The primary challenge was updating the processing script to iterate over multiple files, compute individual periodic totals, and concatenate the DataFrames into a unified dataset for global aggregation without losing performance.
- **Duplicated products:** There were some duplicated products for this task that incremented the total sales significantly.

3. Solutions & Implementation Strategy

- **Technology Stack:** Python with Visual Studio Code (VS Code) inside the established Python virtual environment (`venv`) using `pandas`.
- **Iterative Processing & Concatenation:**
    - Created a list containing the names of the three CSV files.
    - Used a loop to iteratively read each file with `pd.read_csv`, parse numerical values in the `importe` column using `pd.to_numeric`, and compute individual periodic sales totals.
    - Stored each DataFrame in a list and combined them using `pd.concat(lista_dfs, ignore_index=True)` to derive the overall annual revenue and transaction counts in a single global calculation pass.

- **Using drop_duplicates()** to fix the duplication mistakes obtaining the correct results. (`df = df.drop_duplicates(subset="id_venda")`

4. Source Code (`sumV2.py`)

```
import pandas as pd

# 1. Lista con los nombres de tus 3 archivos CSV
archivos = ["vendas_2025_cuadrimestre1.csv",
"vendas_2025_cuadrimestre2.csv", "vendas_2025_cuadrimestre3.csv"]

# Lista para guardar los DataFrames cargados
lista_dfs = []

print("--- RESULTADOS POR CUATRIMESTRE ---")

# 2. Leer y calcular cada trimestre
for i, archivo in enumerate(archivos, start=1):
    # Leer CSV
    df = pd.read_csv(archivo, sep=";", decimal=",")

    # Eliminar ventas repetidas dentro de cada archivo
    df = df.drop_duplicates(subset="id_venda")

    # Asegurar conversión a número
    df["importe"] = pd.to_numeric(df["importe"],
errors="coerce").fillna(0)

    df["periodo"] = i
    # Guardar en la lista para el cálculo global
    lista_dfs.append(df)

# 3. Unir los 3 CSVs y eliminar ventas repetidas entre archivos
df_total = pd.concat(lista_dfs,
ignore_index=True).drop_duplicates(subset="id_venda")

for i in range(1, len(archivos) + 1):
    df_periodo = df_total[df_total["periodo"] == i]
    dinero_periodo = df_periodo["importe"].sum()
    ventas_periodo = len(df_periodo)
```

```
    print(f"Cuatrimestre {i}: {dinero_periodo:.2f} € | {ventas_periodo}
ventas")

total_dinero_global = df_total["importe"].sum()
total_ventas_global = len(df_total)

print("\n--- TOTAL ACUMULADO (3 CUATRIMESTRES) ---")
print(f"Total de dinero recaudado: {total_dinero_global:.2f} €")
print(f"Cantidad total de ventas:  {total_ventas_global}")
```

5. Software & Environment Configuration

- **Integrated Development Environment (IDE):** Visual Studio Code (VS Code)
- **Programming Language:** Python
- **Data Processing Library:** Pandas
- **Environment Isolation:** Python Virtual Environment (`venv`)

6. Bibliography

- **Pandas Documentation:** https://pandas.pydata.org/docs/ - Official reference for Python Pandas, used for multi-file data manipulation, iteration, and concatenation.

7. Usage of AI in the Project

Technologies Used

- **Gemini for Code Generation:** Assisted in writing and updating the Python script (`sumV2.py`) to process multiple CSV files, compute period totals, and aggregate overall annual sales.
- **Gemini for Document Creation:** Formatted, structured, and assembled the project report cleanly into a Google Doc based on updated requirements.
- **Gemini for Prompt Engineering & Improvement:** Refined rough or informal instructions into clear, structured, and effective prompts for improved AI comprehension.

Prompts Used in the Project

1. **Code Generation Prompt (Translated from Spanish):** *"I need to create a program that takes a CSV and can sum the total sales in money and the quantity of sales. How could I do it in the simplest way possible?"*

2. **Prompt Refinement / Improvement Prompt:** *"Improve this prompt while keeping everything in it saying the same things, only for an AI to understand it better and keeping it as brief as it is right now:*

Here is a sketch of our report document, in which we needed to handle a task of counting the total sales of a csv and summing the total ammount of money in them. I want you to create a google docs well formated correcting the grammar and following the same structure as I present it.

You are allowed to change some things , for example I repeat myself a lot in which are errors we had and how we solved them in multiple parts, and mixed it with what the problem we had ahead was."

3. **Document Creation Prompt:** *"Here is a draft of our report on a task involving counting total sales and summing revenue from a CSV file.*
   Please create a well-formatted Google Doc based on this content. Follow the provided structure and correct any grammar errors. You may consolidate repeated sections—specifically where errors, solutions, and problem descriptions are redundant—to improve clarity and flow while keeping all core information intact."
4. **Document Editing & Bibliography Prompt:** *"Edit it, including a brief bibliography which you add links to documentation from python pandas (https://pandas.pydata.org/docs/)"*
5. **Multi-File Script Update Prompt:** *"This time, we have to do three different csv files, so the new challenge is simply to change the script in order to get the summatory of the total sales and the total ammount of sales in the year 2025. The task only asks for the total but we also will share in the document the total of each fourth mode period. Here is the script so you can see the differences. Create another google docs the same as always but with this changes."*